

GLS UNIVERSITY
Faculty of Computer Application &IT
PG Programme - MScIT
Semester - III
Second Mid Term Exam -2025

Subject Name: Working with Network Analysis Tools

Subject Code: 210804333

Date: 11/10/2025

Total Marks: 50

Time: 7:30AM to 8:30PM

Enrollment No: _____

MachineID: _____

SET-1

Instructions: Open the provided set1.pcapng file in Wireshark.

Answer all questions with proper evidence (screenshots or written steps).

1. Apply display filters to isolate only TCP traffic to port 443. Save filtered output to 'filtered443.pcapng'
2. Use filter expressions to find all HTTP GET requests in the file and extract the hostnames.
3. Combine two conditions using logical operators: (ip.src == 192.168.1.2) && (tcp.port == 80). Apply and explain the result
4. Identify and explain the following: TCP ACKed unseen segment, TCP Retransmission, TCP segment not captured. Add screenshots of packet numbers and descriptions.
5. Apply a filter to find all DNS queries for .org domains using regular expressions.
6. Filter all packets that include the word 'login' in HTTP requests.

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SET-2

Instructions: Open the provided set2.pcapng file in Wireshark.

Answer all questions with proper evidence (screenshots or written steps).

1. Create a filter to show all packets from 192.168.10.5 to 172.16.0.8 using TCP.
2. Display Filter Fields to identify and extract the top 5 destination IPs in the capture.
3. Use the Statistics > Protocol Hierarchy feature in Wireshark to generate a breakdown and interpret the usage percentage of HTTP and DNS
4. Create and apply a regular expression filter to find URLs containing 'admin' or 'dashboard' in HTTP requests.
5. Export only the UDP packets to a new file and write steps taken.
6. Identify TCP Duplicate ACKs and TCP Previous Segment Not Captured. Explain how these impact the transmission and include packet screenshots.

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SET-3

Instructions: Open the provided set3.pcapng file in Wireshark.

Answer all questions with proper evidence (screenshots or written steps).

1. Filter: Source IP 10.0.0.5 and destination port 53 (DNS).
2. Use membership operator to extract packets involving ports: 80, 443, 21.
3. Use Follow TCP Stream to analyze one HTTP file transfer.
4. Create a filter to find packets containing 'login' or 'password' in payload.
5. Export packets for a single TCP stream and count the total number of packets.
6. Analyze and identify TCP port reuse, impact on session, and provide explanation with packet numbers.

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SET-4

Instructions: Open the provided set4.pcapng file in Wireshark.

Answer all questions with proper evidence (screenshots or written steps).

1. Apply regex filter to find HTTP requests with .php, .html, .js extensions.
2. Combine filters to show DNS queries from 192.168.1.10 and TCP packets to port 443.
3. Save TCP packets from 192.168.1.2 to 192.168.1.5 to a new file and describe the filter used.
4. Extract UDP traffic with size > 100 bytes.
5. Find top 5 TCP conversations by bytes.
6. Identify TCP segment retransmissions, Keep-Alive and Keep-Alive ACK packets. Explain their use.